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How environmental knowledge and concern shape Chinese consumers' green purchase intention?

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ABSTRACT

The planet experiences a sense of urgency in environmental issues, and the economic growth of China accelerates the pressure on sustainable consumption. More frequent and extreme weather events and climatic variation are threatening the safety of people and food security in China. Hence, it is essential to understand the people's attitudes and behaviors towards environmental protection. Drawing on a Theory of Planned Behavior (TPB)-informed integrated model, this paper investigates how environmental concern (EC) and environmental knowledge (EK) influence green purchase intention (GPI) among Chinese consumers, with subjective norms (SN), perceived behavioral control (PBC), and green perceived value (GPV) examined as mediating mechanisms. The study addresses a more specific gap in the literature by examining whether social, control, and value-based pathways differently mediate the effects of EC and EK on GPI within one model. The survey involved 546 consumers and data was analyzed through SmartPLS 4.0. The findings reveal that EC and EK positively influence SN and PBC. SN and PBC mediate the effects of both antecedents on GPI, whereas GPV does not emerge as a significant mediator. The direct effect remains significant for EC but not for EK, indicating partial mediation for EC and indirect-only influence for EK. These findings indicate that SN and PBC are the principal pathways through which EK and EC translate into GPI, while GPV does not function as a significant mediating route in this sample. This paper offers an initial empirical basis to further studies and is consistent with the aims of the United Nation's sustainable development goal (SDG) 12 relating to responsible consumption and production and SDG 13, which is to promote climate action.

1. Introduction

The long-term economic progress of China during the past few decades has greatly enhanced living standards and income levels, but it has been associated with significant environmental costs. Economic affluence has increased resource-consuming and producing behaviors that go beyond the ecological thresholds, causing extensive environmental destruction and rising climate-related hazards. Increasing temperatures around the world have led to the loss of biodiversity, sea-levels rising because of melting polar ice, more common severe weather phenomena and ocean acidification which together leads to the destabilization of important ecological systems (Jing and Rasiah, 2025). Pollution is one of the greatest challenges to life expectancy in China, where the level of pollution is estimated to be six times higher than suggested by the World Health Organization (Li et al., 2025). This issue of contamination and environment degradation arises due to the growth in the economy and the level of productivity, and its urgency increases (Almohammadi & Abdulghaffar, 2022). All these issues underscore the importance of promoting responsible consumption and production (Sustainable Development Goal (SDG) 12) and enhancing climate action (SDG 13). Governments, communities, corporations and personal citizens are laying more emphasis on the problem of ecology and environment protection (Amena, 2023; Kaiser et al., 1999), while constantly developing environmental policies to inspire and drive people to take care of

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the environment (Jaiswal & Kant, 2018). Under this context, the notion of green purchase intention (GPI) can be regarded as critical since consumer purchasing behavior is one of the essential mechanisms of their behavior through which sustainable consumption can be promoted and environmental pressure alleviated (Ogiemwonyi, 2024) (see Table 1).

As a reaction to the pressures of the environment, companies are also beginning to take up green measures, including manufacturing environmentally safe products and packaging (Imiru, 2023; Dunlap et al., 2000), whereas consumers are moving to greener living and focusing on buying green goods (De Gregorio et al., 2025; Amena, 2023; Ahn and Kwon, 2019; Li et al., 2021; Rasiah et al., 2023), since air contamination and carbon emissions are among the most obvious environment problems to be experienced by consumers (Zhang et al., 2024). As purchasing behavior directly influences firms' production choices and broader environmental outcomes (Amoako et al., 2020; Zheng et al., 2020), understanding consumers' GPI is therefore critical, motivating this study's focus.

While research on GPI has received increasing attention (Imiru, 2023; Rasiah et al., 2023; Almohammadi & Abdulghaffar, 2022; Naz et al., 2020), there exists several gaps in the literature.

First, a research gap exists regarding the direct influence of environmental concern (EC) on green purchase intention (GPI). Although previous studies have highlighted the paradox between increasing environmental concern and continued overconsumption, they generally report a positive relationship between EC and sustainable purchase intentions. However, most studies based on the Theory of Planned Behavior (TPB) have examined EC primarily as an antecedent of attitude rather than as a direct predictor of behavioral intention (Chen et al., 2021). Consequently, the direct effect of EC on GPI remains insufficiently explored, and this study addresses this gap by examining whether EC directly influences GPI.

Second, a critical research gap remains in understanding the influence of environmental knowledge (EK) on green purchase intention (GPI). Although sustainable consumption has received increasing scholarly attention, existing evidence suggests that many consumers continue to possess limited knowledge of green products (ElHaffar et al., 2020; Kamalanon et al., 2022), while widespread misinformation further undermines informed purchasing decisions (Khan et al., 2020). These persistent knowledge deficiencies raise questions about whether EK effectively translates into green purchase intention, indicating that its predictive role remains inadequately established and warrants further empirical examination. Besides the above-mentioned gaps, there have also been an array of contradictory findings in the extant literature leading to confusion-spotting gaps. Despite TPB-based assumptions that subjective norms (SN) influence GPI (Qi & Ploeger, 2021; Kamalanon et al., 2022; Lavuri, 2021), empirical evidence remains mixed, with many studies reporting non-significant effects (Hamzah & Tanwir, 2021; Kamalanon et al., 2022; Kumar & Mohan, 2021; Rausch & Kopplin, 2021; Rizkalla et al., 2022) and others finding significant relationships (Qureshi et al., 2022; Rizkalla et al., 2022; Rashid & Lone, 2023), highlighting an unresolved gap in the literature. Although perceived behavioral control (PBC), GPV, EK, and EC are theoretically linked with GPI (Lavuri, 2021; Kumar & Mohan, 2021; Zhuang et al., 2021; Li et al., 2021), empirical evidence remains inconsistent, with several studies reporting non-significant or mixed effects (Zahan et al., 2020; Sinha & Annamdevula, 2023; Rizkalla et al., 2022), while most studies have highlighted direct relationships and largely overlooked the mediating mechanisms through which these factors influence GPI (Sinha & Annamdevula, 2023; Testa et al., 2024).

Table 1
Demographic profile

Demographic items	Frequency	Percentage (%)
Age		
≤18	8	1.5
19-23	485	88.8
24-28	28	5.1
29-32	6	1.1
≥33	19	3.5
Gender		
Female	363	66.5
Male	183	33.5
Education		
Junior College Pre-University	50	9.2
Degree	463	84.8
Masters	15	2.7
PhD	18	3.3

While research on GPI has expanded considerably, important issues remain unresolved. Prior studies report mixed evidence on whether EK and EC influence GPI directly, indirectly, or not at all. In addition, the mechanisms through which these distal environmental orientations are translated into purchase intention are rarely examined simultaneously. In particular, limited attention has been given to whether SN and PBC function as social and control pathways, and whether green perceived value (GPV) functions as a value-based evaluative pathway, within the same model in the Chinese context. Accordingly, this study does not claim a full extension of the canonical Theory of Planned Behavior (TPB). Rather, it develops a TPB-informed integrated model that retains SN and PBC as proximal TPB mechanisms, treats EK and EC as distal antecedents informed by the Value-Belief-Norm perspective, and examines GPV as a context-specific evaluative construct grounded in perceived value theory. The contribution therefore lies in comparatively testing multiple mediating pathways linking EK and EC to GPI, rather than in claiming geographical scarcity of prior research. This study's theoretical contribution lies in documenting a mediating pathway pattern in this sample: SN and PBC are stronger mediating channels than GPV in linking EK and EC to GPI. This comparative mediating-pathway finding contributes to the literature by clarifying that, in a collectivist market context, social and feasibility-based mechanisms are more proximate drivers of green intention than product-value appraisal alone.

We emphasize that this finding extends rather than disrupts existing TPB-based green consumption literature: the dominance of normative and feasibility mechanisms overvalue appraisal in collectivist societies is theoretically anticipated. The study's contribution therefore lies not in establishing a disruptive or boundary-testing theoretical claim, but in offering a comparative empirical assessment of three simultaneously modeled mediating pathways (SN, PBC, GPV) within a TPB-informed integrated model in the Chinese context, using a sufficiently powered dataset (n = 546). This comparative pathway-level evidence contributes to the green consumption literature by demonstrating the relative salience of normative and control-based mechanisms vis-a-vis value appraisal, a distinction that prior studies have often examined in isolation rather than jointly (Kamalanon et al., 2022; Lavuri, 2021; Sinha & Annamdevula, 2023). Importantly, this study is presented as a carefully scoped comparison of selected mediating mechanisms within the Chinese context.

2. Literature review

2.1. Theoretical review

The TPB is a highly organized model that is used to explain behavioral intentions (Lavuri, 2021), and it is commonly applied across different contexts (Amoako et al., 2020; Qi and Ploeger, 2021). TPB is used as a tool to examine SN and PBC as underlying factors and potential mediators of GPI which broadens the scope to the more recent area of sustainable consumption and also fills the gap in the understanding of the psychological and social factors of environment-friendly behavior.

According to the TPB, behavioral intention is proximally determined by attitude toward the behavior, SN, and PBC (Ajzen, 1991, 2011). In the present study, SN and PBC are retained from the TPB framework.

However, the general attitude construct is not modeled separately. Instead, GPV is examined as a narrower, context-specific evaluative appraisal of green products. GPV is therefore not treated as theoretically synonymous with attitude; rather, it captures consumers' assessment of the net benefits and sacrifices associated with green purchases, consistent with perceived value theory (Zeithaml, 1988; Chen & Chang, 2012).

This modeling choice is appropriate for green purchase decisions because such decisions often involve explicit trade-offs related to price, quality, credibility, and effort. At the same time, because attitude is a core TPB construct and has repeatedly shown strong effects on intention in prior syntheses, the present framework should be interpreted as TPB-informed rather than as a full canonical TPB model (Armitage & Conner, 2001; Zaremohzzabieh et al., 2021). Related green consumption research has also typically retained attitude when integrating value-based constructs with TPB, rather than treating the two as identical (Wang et al., 2022). Accordingly, what is lost by omitting general attitude is the ability to compare broad evaluative disposition toward green purchasing with product-specific value appraisal; this is acknowledged as a limitation and an avenue for future research. Critically, this design choice rests on a principled construct-level distinction: attitude in the canonical TPB refers to a global evaluative disposition toward the behavior, whereas GPV captures a narrower, product-specific cost–benefit appraisal grounded in perceived value theory (Zeithaml, 1988). These constructs differ in theoretical origin, referent scope, and empirical operationalization. Including both simultaneously in future research would allow a direct test of whether broad evaluative disposition and product-specific value appraisal function as complementary or redundant pathways to GPI. The present study treats GPV as a context-appropriate evaluative construct while explicitly acknowledging that the omission of general attitude prevents a complete canonical TPB test.

The present study integrates three complementary lenses. First, the Value-Belief-Norm (VBN) perspective helps explain why pro-environmental orientations emerge by emphasizing the role of environmental beliefs and values; accordingly, EK and EC are treated here as distal antecedents (Stern et al., 1999; Oreg & Katz-Gerro, 2006). Second, TPB explains the proximal mechanisms through which these orientations are translated into intention, particularly via SN and PBC (Ajzen, 1991, 2011). Third, perceived value theory specifies the evaluative route through which consumers assess the benefits and sacrifices of green products (Zeithaml, 1988). In this way, the model links distal environmental orientations (EK and EC) to proximal social and control mechanisms (SN and PBC) and to a product-specific evaluative appraisal (GPV). The three frameworks are not merely stacked but integrated through an explicit causal sequencing: VBN-derived constructs explain why environmental orientations form (i.e., through accumulated knowledge and affective concern); TPB mechanisms specify how these orientations are channeled into behavioral intention through SN, as the perceived social-expectation pathway, and PBC, as the perceived-feasibility pathway; and perceived value theory provides an alternative evaluative route through GPV, whose comparative weakness reveals a theoretically meaningful tension. Specifically, GPV and SN differ fundamentally in their locus of motivation: GPV reflects individual cost–benefit calculation, whereas SN reflects the perceived social obligation to conform to environmental norms. In high-collectivism contexts, the latter may carry greater motivational force, helping to explain the stronger role of SN than GPV in the present model. This theoretical architecture means that each framework carries a distinct and non-redundant explanatory function: VBN-derived constructs (EK, EC) would have limited explanatory scope without TPB mechanisms to specify their behavioral translation, and the comparative weakness of the GPV pathway would be theoretically uninterpretable without perceived value theory to ground its conceptual role.

2.2. Empirical review

2.2.1. EK and GPI

EK refers to an individual's thorough comprehension and awareness of ecological issues, including their underlying reasons and complex consequences (Mayangsari & Fabiola, 2020; Cui et al., 2024). This knowledge base is regarded as a critical precursor and a foundation upon which other influential factors are built (Zahan et al., 2020; Fryxell and Lo, 2003). A shortage of EK, may perpetuate apathy, inaction, and unsustainable consumption patterns, emphasizing the critical effect of environmental education and awareness campaigns in facilitating the conversion into a more eco-conscious society (Kennedy & Adhikari, 2022; Mayangsari & Fabiola, 2020). Simultaneously, research has shown that consumers' propensity to partake in environmental-friendly purchasing behavior is affected by their level of ecological knowledge (Rizkalla et al., 2022). EK affects consumers' intentions to participate in environmental-friendly purchasing behavior (Rehman & Harun, 2022). According to research, there is a clear correlation between a person's comprehension of the environment and their proclivity to make environmentally conscious purchasing decisions (Kennedy & Adhikari, 2022). The research manifested a positive and key association between a person's EK and his intention to make environmentally conscious purchasing decisions (Mayangsari & Fabiola, 2020). Consumers with greater environmental knowledge (EK) are more likely to prioritize sustainability when making their purchasing decisions (Zahan et al., 2020; Li et al., 2023).

H1. EK has a significant relationship with GPI

2.2.2. EC and GPI

EC represents an individual's level of consciousness, interest, and dedication to environment protection and conservation (Kumar & Mohan, 2021). It encompasses a person's concern and engagement in environmental issues, reflecting their perceived urgency to address the negative effects of human activities on ecological systems (Mayangsari & Fabiola, 2020). Environmentally conscious people are anticipated to partake in actions benefiting the planet, like a strong preference for eco-friendly or "green" products and services (Rausch & Kopplin, 2021). EC is thought to contribute to a stronger perception of green value among consumers by increasing their appreciation for the ecological benefits of environmentally conscious products and services (Zhuang et al., 2021). Since there is an increase in human awareness of the complex relationships among the manner of consumption, the degradation of the environment, and solutions to these issues, they will probably have a more profound understanding of how to identify the underlying values of green products (Li et al., 2021). The additional knowledge and understanding may cause a higher readiness to spend more on those items because the customers are valuing the environment-related advantages they get (Rasiah et al., 2023).

H2. EC has a significant relationship with GPI

2.2.3. The mediating role of SN

The concept of GPI has been widely studied as a critical antecedent of sustainable consumption behavior. New applications of the Theory of Planned Behavior remain focused on the central position of SN, which is the perception of social influence of significant others to participate in environmentally friendly behaviours, in determining green consumption intentions (Ajzen, 2011). In green consumption contexts, SN captures the degree to which consumers perceive social approval or expectations regarding the purchase of environmental-friendly products. Recent empirical studies manifest that EK enhances individuals' awareness of socially endorsed environmental values and sustainability-related expectations, thereby strengthening SN associated with green purchasing (Li et al., 2021). Well-informed consumers tend more to recognise that environmentally responsible consumption is socially valued, which indirectly increases their GPI through heightened

normative pressure. This suggests that SN serves as an important social–psychological mechanism linking EK to GPI. Similarly, EC, reflecting individuals' level of worry and moral engagement with environmental issues, has been exhibited to indirectly affect GPI by reinforcing perceived social expectations for pro-environmental behavior (Wang et al., 2022; Khan et al., 2023). Those who are environmentally conscious tend more to be more susceptible to normative signals and social talk on sustainability and buy along what they assume is considered socially acceptable. It has been recently reported again by studies that SN are the main mediators of EC converting it into GPI especially in socially embedded consumption environments (Chen et al., 2021).

When combined, the recent literature indicates that SN are a very important mediating channel along which EK and EC affect GPI. In turn, this research puts forward these hypotheses:

H3. SN mediates the relationship between EK and GPI.

H4. SN mediates the relationship between EC and GPI.

2.2.4. The mediating role of PBC

GPI is considered a proximal predictor of sustainable consumption behavior. The term PBC in the TPB describes individual perceptions of their ability and resources to participate in a specific behavior which may be either an internal control (e.g. skills and knowledge), or an exterior constraint (e.g. time, cost, and accessibility) (Ajzen, 2011). In the green consumption setting, PBC is about how easy or hard it is perceived by consumers to purchase environmental-friendly products. The recent findings indicate that EK can boost the feeling of controlling green purchasing decisions due to its contribution to the improvement of consumer competencies in recognizing green products, assessing environmental claims, and making sustainable consumption choices (Li et al., 2022; Chen and Liu, 2023). As individuals become more knowledgeable about EC and green products, they are more likely to perceive fewer barriers and greater self-efficacy in engaging in green purchasing. Consequently, EK may indirectly influence GPI by strengthening PBC. Similarly, EC, which reflects individuals' level of awareness and emotional involvement with environmental problems, has been found to increase motivation to overcome perceived barriers to pro-environmental behavior (Wang et al., 2022; Khan et al., 2023). Environmentally concerned consumers tend to invest more effort in seeking green alternatives and are more content to adjust their consumption habits, thereby enhancing their PBC over green purchasing. Empirical evidence indicates that PBC functions as an important psychological mechanism through which EC is translated into GPI, particularly in contexts where green consumption involves higher perceived costs or effort (Liu et al., 2023).

Overall, recent literature supports the mediating role of PBC in linking both cognitive (EK) and affective (EC) environmental factors to GPI. PBC is one of the main ways in which environmental orientations are transformed into intentional green purchases because it enhances the perception of control and feasibility among individuals. In line with this, the subsequent hypotheses have been raised:

H5. PBC mediates the relationship between EK and GPI.

H6. PBC mediates the relationship between EC and GPI.

2.2.5. The mediating role of GPV

Green perceived value (GPV) represents a key cognitive evaluation mechanism that has received increasing scholarly attention. Recent studies suggest that GPV plays a pivotal role in translating individuals' environmental orientations into green purchase intention (GPI). The concept of GPV denotes consumers overall assessment of the net benefits of green products with regard to environmental, functional, emotional, and at times social values versus the perceived cost of those benefits (Sangroya and Nayak, 2017). It is demonstrated that EK enhances the capacity of consumers to identify and assess the environmental and functional characteristics of green products, which increases their perceived green value. People with high EK trend more to tell real green products apart and be able to enjoy both the environmental and personal advantages of green consumption (Li et al., 2022). Therefore, EK is able to affect GPI indirectly by enhancing GPV, which is a proximal predictor of purchase-related decisions. Likewise, EC has also been identified to have an effect on consumers value based judgments of green products due to the fact that EC represents individual awareness and emotional engagement with ECs. Green value perception is positively influenced by environmentally conscious consumers who are inclined to give weight to environmental benefits and moral fulfillment when evaluating product value and thus can perceive green value regardless of higher price or performance uncertainty (Huang et al., 2024). As indicated by the latest empirical findings, GPV might act on the mediating effect of EC into GPI because it matches environmental motivations with positive value perceptions among consumers.

To sum up, the current literature (DeJun et al., 2025; Zhang et al., 2024) also confirms the idea that GPV is an essential intermediary that can be used to connect EK and EC with GPI. The combination of cognitive understanding and affective concern to form a unified value appraisal enables GPV to convert environmental orientations into intentional green purchasing behavior. As such, this paper will offer the below hypotheses:

H7. GPV mediates the relationship between EK and GPI.

H8. GPV mediates the relationship between EC and GPI.

This paper has a research framework as shown in Fig. 1 that has been informed by critical reviews of theoretical and empirical literature (see Fig. 2).

3. Methodology

This section explains the methodology that can be empirically used in testing the hypotheses that are developed according to the literature review.

3.1. Research design and sampling

A cross-sectional research design and a convenience sampling technique were used to conduct an online questionnaire survey, a research design that is often used in consumer-oriented sustainability research. Volunteers were recruited with the help of easily available online sources, which would limit the representativeness of volunteers to the whole population. The research considers consumers within mainland China where the situation can be described as a long-term presence of state-managed environmental programs, which has also led to increased interest among the population in environmental problems. Consequently, people in this environment tend to highlight the environment. The data were gathered through an online investigation and 546 responses were found to be valid. It comprised of 363 females and 183

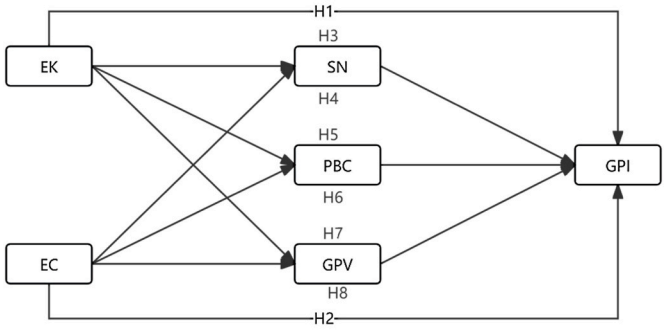


Fig. 1. Research framework.

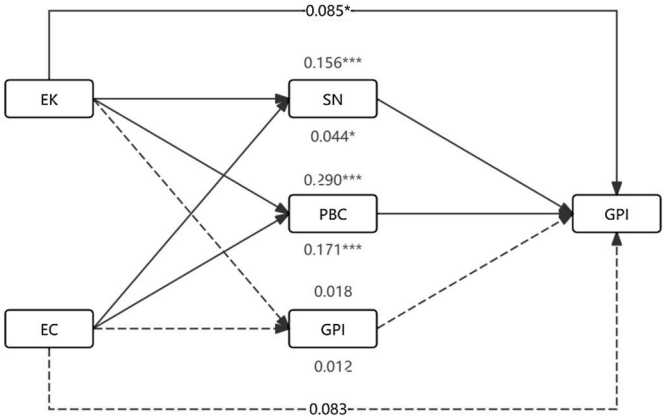


FIGURE 2. Structural model.

males and most of them belonged to 19-23 years age group (88.8%). Most of the participants had been educated at the level of a university (84.8%).

3.2. Data and analysis

The questionnaire which is to be used in this quantitative research was designed through the modification of tested measurement scales which have been applied to prior similar research. Purchase intention about green (GPI) was determined according to the items modified by Akbar et al. (2014) and Kim et al. (2013). Three items based on Chan and Lau (2002) were used for determining SN and three items based on Kim et al. (2013) to measure PBC. Values on GPV were based on Chaudhuri (1997). Five items based on Mostafa (2007) were used to evaluate both EK and EC. All these validated measures made up an integrated and theoretically-based questionnaire. Structural equation modeling with Smart PLS 4.0 was the main tool of data analysis in this study. SmartPLS was applied to determine the relevance between the constructs and to test the predictive ability of the mode. PLS-SEM was considered appropriate owning to its suitability for intricate structural models and analytical efficiency. Furthermore, the extensive use of PLS-SEM in prior studies on sustainable and green consumption supports its applicability for model evaluation in this study (Armutcu et al., 2024; Pandey & Yadav, 2023). PLS-SEM was selected over CB-SEM due to the research's predictive focus and composite-based model structure. Empirical evidence from Dash and Paul (2021) shows that PLS-SEM outperforms CB-SEM in indicator loadings, AVE, CR, and explained variance (R^2), making it more suitable for predictive research despite CB-SEM's superiority in global fit indices. It is explicitly acknowledged that CB-SEM would have been equally appropriate for the theory-confirmatory components of this study, and its use in future replication is strongly recommended to verify model fit indices. The choice of PLS-SEM in the present study rests on three specific considerations: first, the model integrates constructs drawn from three distinct theoretical traditions (TPB, VBN perspective, perceived value theory), making composite-based estimation more aligned with the multi-source theoretical structure; second, the study's primary inferential interest lies in the relative magnitude and significance of competing indirect effects, for which PLS-SEM's bootstrapping procedures provide well-validated inference; and third, the emphasis on explained variance (R^2) in GPI as a benchmark of pathway importance complements the coefficient-level analysis. These considerations justify the PLS-SEM choice while acknowledging its limitations relative to CB-SEM for global fit assessment.

To conduct the analysis, both measurement and structural models were evaluated and compared to the data. The measurement model evaluated internal consistency and validity (construct) through Cronbach's alpha and CR (composite reliability), with a threshold of 0.7 established for each of these characteristics. A CR exceeding 0.7 and an AVE (Average Variance Extracted) exceeding 0.5 were used to assess the study's convergent validity. The discriminant validity of the test was evaluated through the Fornell-Larcker criterion, created by Fornell and Larcker in 1981. According to this criterion, the AVE for each construct must surpass the relationships between it and other constructs.

4. Findings

The study first evaluated the measurement model. Indicator reliability was evaluated by means of standardized outer loadings, with values surpassing the recommended threshold of 0.708, indicating that each construct explains over 50% of the variance in its indicators (Hair et al., 2019, 2021). In this study, all measurement items satisfied this criterion except EK1, EC5, SN1, GPV3, GPV4, and GPI4, which were removed due to insufficient loadings. This item removal corresponds to prior research. This illustrates that indicators with weak loadings may compromise construct validity and overall model fit (Razali et al., 2023). Scores of Cronbach alpha (0.871-0.95) and CR (0.912-0.967) are high indexes that indicate that all the constructs are highly consistent and credible internally. The scores on AVE are another indicator of convergence among the constructs (ranging between 0.721 and 0.937).

The dimensions all exhibited that Cronbach's alpha values exceeded 0.7, high CR values over 0.7, and large AVE values over 0.5. The empirical evidence supports the convergent validity of the constructs.

As it can be seen in Table 3, the square roots of AVE values of each construct exceed the relevance between other constructs, which means that there is discriminant validity in light of the Fornell and Larcker criterion (1981) (see Table 2). To investigate the likely existence of common method bias in the sample, Harman's single-factor test was performed. In particular, all the measurement items applied to the research were analyzed through the unrotated exploratory factor analysis in SPSS to identify if it can be explained by a single latent variable as the main part of the overall variance. It turned out that the first factor explained 45.505 percent of the variance, which is less than the 50 percent threshold recommended by Podsakoff et al. (2003). Accordingly, these results suggest that common method bias is unlikely to fully account for the observed relationships, although it cannot be ruled out given the single-source, cross-sectional design. To complement and partially address the recognized limitations of Harman's single-factor test, full collinearity variance inflation factor (VIF) scores were computed for all constructs following the procedure recommended by Kock (2015) for PLS-SEM contexts. All VIF values ranged between 1.543 and 3.291, below the conservative threshold of 3.3, providing additional but not conclusive evidence against severe common method bias. Future single-source studies should additionally employ marker variable procedures to provide a more rigorous safeguard. Nevertheless, these statistical tests have recognized limitations for studies relying entirely on self-reported perceptual measures. Future studies should employ procedural remedies such as temporal separation of predictor and outcome measures, inclusion of a marker variable, or use of the unmeasured latent method factor (ULMF) approach (Lindell & Whitney, 2001; Podsakoff et al., 2003) to provide stronger protection against common method variance.

Table 2
Assessment of measurement model

	Cronbach's alpha	CR	AVE	R-square
EC	0.871	0.912	0.721	
EK	0.950	0.964	0.870	
GPI	0.941	0.958	0.850	0.745
GPV	0.907	0.942	0.843	0.642
PBC	0.876	0.924	0.802	0.665
SN	0.932	0.967	0.937	0.609

Table 3
Discriminant validity

	EC	EK	GPI	GPV	PBC	SN
EC	0.849					
EK	0.567	0.933				
GPI	0.620	0.722	0.922			
GPV	0.666	0.745	0.711	0.918		
PBC	0.664	0.767	0.843	0.768	0.895	
SN	0.559	0.766	0.785	0.734	0.823	0.968

We explicitly acknowledge that the procedural remedies employed here—Harman's single-factor test and full collinearity VIF—provide important but imperfect safeguards against common method variance. All constructs in this study are perceptual and self-reported, collected from a single source at a single time point, using the same response scale and questionnaire format. More rigorous procedural remedies, such as temporal separation of predictor and criterion measurement, inclusion of a theoretically unrelated marker variable (Lindell & Whitney, 2001), or statistical techniques such as the unmeasured latent method factor (ULMF) approach (Williams et al., 2010) and CFA-based marker variable techniques, would have provided substantially stronger protection against CMB. We recommend these as mandatory design features for future replications of this model.

After validating the convergent and discriminant validity of the measurement model, the focus shifted to evaluating the structural model and empirically validating the hypotheses proposed.

The study's model fit was acceptable, as the SRMR value of 0.059 was below the recommended threshold of 0.08. Table 4 shows that the total effects of both EC and EK on GPI are positive and significant. However, the direct-effects model presents a more differentiated pattern. EC retains a weak but statistically significant direct effect on GPI ($\beta = 0.085$, $p = 0.036$), whereas the direct effect of EK on GPI is not significant ($\beta = 0.083$, $p = 0.162$). This suggests that the influence of EK on GPI is indirect, while the influence of EC on GPI is both direct and indirect. With regard to the indirect effects, EK exerts a significant positive influence on GPI through SN ($\beta = 0.156$, $t = 3.394$, $p = 0.001$) and through PBC ($\beta = 0.290$, $t = 7.080$, $p < 0.001$), supporting H3 and H5. However, the indirect effect of EK on GPI through GPV is not significant ($\beta = 0.018$, $t = 0.594$, $p = 0.553$); therefore, H7 is rejected. Similarly, EC exerts a significant indirect influence on GPI through SN ($\beta = 0.044$, $t = 2.436$, $p = 0.015$) and through PBC ($\beta = 0.171$, $t = 5.910$, $p < 0.001$), supporting H4 and H6. By contrast, the indirect effect of EC on GPI through GPV is not significant ($\beta = 0.012$, $t = 0.595$, $p = 0.552$); therefore, H8 is rejected. Taken together, these findings indicate partial mediation for EC and indirect-only influence for EK, with SN and especially PBC serving as the strongest mediating pathways.

Table 4
Structural model assessment

Path	Coefficients	Standard deviation (STDEV)	T statistics |O/STDEV|	P values	Decision
Total Effect Model					
EC - > GPI	0.311	0.048	6.501	0.000	Supported
EK - > GPI	0.546	0.039	13.944	0.000	Supported
Direct Effects					
EC - > GPI	0.085	0.040	2.100	0.036	Supported
EK - > GPI	0.083	0.059	1.398	0.162	Not Supported
Indirect Effects					
EC - > SN - > GPI	0.044	0.018	2.436	0.015	Supported
EC - > PBC - > GPI	0.171	0.029	5.910	0.000	Supported
EK - > SN - > GPI	0.156	0.046	3.394	0.001	Supported
EC - > GPV - > GPI	0.012	0.020	0.595	0.552	Not Supported
EK - > PBC - > GPI	0.290	0.041	7.080	0.000	Supported
EK - > GPV - > GPI	0.018	0.030	0.594	0.553	Not Supported

Symbols used in the table are ✓ (for Supported), and × (for Rejected).

5. Discussion

This research delves into the mechanisms through which EK and EC influence Chinese consumers' GPI, incorporating the mediating effect of SN, PBC, and GPV. Our results offer insights that both align with and extend the established literature on green consumer behavior.

To begin with, the total effects indicate that both EK and EC are important antecedents of GPI. However, the decomposition of these effects reveals a more nuanced pattern. The direct effect of EK on GPI is not statistically significant, whereas the direct effect of EC remains significant but weak. This suggests that EK influences GPI primarily through indirect pathways, while EC influences GPI through both direct and indirect routes. Accordingly, the relationship between environmental antecedents and GPI in this study is better understood as mediated rather than purely direct. Among the mediating mechanisms, PBC emerges as the strongest pathway. The indirect effects of EK and EC through PBC are both positive and significant, indicating that EK and EC are translated into GPI largely by strengthening consumers' perceptions of capability, feasibility, and control over green purchasing decisions. This is consistent with TPB-informed green consumption research, which emphasizes the importance of perceived control when behaviors are associated with cost, effort, or accessibility constraints.

The most important mediating channels were via PBC. The indirect influence of EK on PBC was strong and meaningful (=0.290), as well as the indirect effect of EC on PBC (=0.171). This indicates that knowledge and concerns are converted into intentions mainly because they improve the perception of power and control of individuals speaking of purchasing green items. It conforms to the fundamental principles of the TPB and the findings reported by Kamalanon et al. (2022) and Lavuri (2021), who highlighted the significance of the PBC in green consumption particularly when the behavior may be accompanied by a perceived constraint such as increased cost or low availability.

SN also functions as a significant mediating pathway for both EK and EC, although its effect is weaker than that of PBC. The stronger indirect effect from EK through SN suggests that knowledgeable consumers may be more responsive to socially endorsed environmental expectations. This pattern is consistent with the view that, in socially embedded consumption contexts such as China, normative influence remains an important channel through which pro-environmental orientations are converted into purchase intention.

The dominance of SN over GPV in our findings can be further understood by engaging with three theoretically motivated contextual mechanisms specific to China. First, from a cultural theory perspective, China scores high on Hofstede's collectivism dimension, where the perceived social expectations of in-group members exert greater motivational force than individual cost–benefit calculations (Hofstede, 2001; Triandis, 1995). In such high-collectivism contexts, normative compliance is not merely a social preference but a deeply internalized behavioral standard, which helps to explain why socially referenced mechanisms (SN) consistently outperform individual evaluative reasoning (GPV) in predicting pro-environmental intentions. Second, China's state-directed environmental governance structure, including the national 'Ecological Civilization' (shengtai wenming) policy agenda and compulsory environmental education programs, has systematically reinforced social norms around environmental responsibility and amplified the normative salience of green behavior across the population (Zhang et al., 2024). These institutional forces have created a normative environment in which environmental responsibility is not only socially encouraged but also publicly endorsed by state and media

discourse. Third, digitally mediated normative influence through platforms such as WeChat, Douyin, and Weibo has rapidly diffused sustainability discourse, creating social environments where environmentally responsible consumption is publicly visible and peer-endorsed—further intensifying the normative pressure experienced by consumers, particularly younger cohorts who are intensive users of these platforms (Chen et al., 2021; Armutcu et al., 2024). Consequently, China functions not merely as a geographic setting but as a theoretically relevant context in which institutional and cultural features may help explain why normative mechanisms appear more salient than value appraisal in this sample.

Contrary to the proposed expectations, GPV did not significantly mediate the effects of either EK or EC on GPI. The non-significant indirect effects ($\beta = 0.018$ and 0.012, respectively) suggest that the utilitarian and hedonic value perceptions of green products did not act as a cognitive bridge in this model. This is a notable departure from studies that identified perceived value as a pivotal driver (Ng et al., 2024; Zheng et al., 2020). A potential explanation lies in the "attitude-behavior gap" frequently visualized in green consumption (Rausch & Kopplin, 2021). Chinese consumers may possess knowledge and concern, and even acknowledge the abstract value of green products, but this valuation may not sufficiently overcome primary barriers like price sensitivity or skepticism (Sinha & Annamdevula, 2023; Zahoor Ur Rehman and Amran, 2022), thus failing to translate into a stronger intention through this particular pathway. From a perceived value theory standpoint, this null result is theoretically informative rather than merely anomalous. It suggests that value appraisal (GPV) is motivationally insufficient to bridge environmental orientations and behavioral intention when structural barriers—such as premium pricing, limited availability, and credibility skepticism—suppress the value-to-intention pathway. This pattern suggest that GPV's mediating role may be context-dependent rather than universal: its mediating role may require threshold conditions, including low price sensitivity, high institutional trust, and well-developed green product markets. Future research should test GPV's mediation under these conditions to determine whether the construct is conceptually misplaced in the present framework or whether its activation depends on contextual prerequisites not present in this sample.

However, we acknowledge that the above interpretation is primarily post hoc and speculative, as these contextual factors were not empirically tested in the present study. Three additional interpretive considerations merit explicit discussion. First, regarding construct positioning: GPV's non-significance may partly reflect the absence of general evaluative attitude from the model. As discussed in the theoretical framework section, attitude in the canonical TPB captures a broader dispositional evaluation of the behavior, whereas GPV captures a narrower, product-specific cost–benefit appraisal. It is possible that GPV has limited motivational force when the broader evaluative disposition toward green behavior is unrepresented in the model—a possibility that cannot be ruled out with the present design and that constitutes a rival explanation for the null result. Second, regarding measurement-level concerns: the GPV measurement in this study drew on Chaudhuri's (1997) general perceived value scale, which may not fully capture the multi-dimensional structure of GPV (environmental, functional, emotional, and social value dimensions) as conceptualized in more recent green-specific instruments (Chen & Chang, 2012; Sangroya & Nayak, 2017). Future research should employ richer, domain-specific GPV measures to determine whether the null result is instrument-dependent. Third, regarding contextual moderating conditions: while the structural barriers discussed above (price sensitivity, credibility skepticism, market immaturity) provide plausible contextual explanations, they remain untested hypotheses. We propose that GPV's mediating role may be activated under conditions of low price sensitivity, high institutional trust, and mature green markets—conditions that future studies can test directly through moderated mediation designs.

From a broader perceived value theory perspective, the non-significant GPV mediation in this study raises an important question: under what conditions does product-level value appraisal become a motivational driver of GPI? If future studies confirm that GPV's mediating function weakens in contexts where normative mechanisms are especially salient, perceived value theory may benefit from more explicit attention to contextual moderators such as cultural orientation, institutional trust, and green-market maturity. This would represent a meaningful theoretical extension of perceived value theory beyond its current focus on individual-level cost–benefit calculus.

Theoretically, this study contributes in multiple ways. First, it shows that the effects of EK and EC on GPI are transmitted more strongly through SN and PBC than through GPV in this sample. This refines TPB-informed green consumption research by indicating that, in the Chinese context, social expectations and perceived feasibility are more salient pathways than value appraisal alone. Second, the results support the usefulness of integrating distal environmental antecedents drawn from the Value-Belief-Norm perspective with proximal TPB mechanisms, thereby offering a more differentiated explanation of how EK and EC are translated into GPI. Fourth, the disappearance of EK's direct effect once mediators are introduced is theoretically meaningful: it suggests that EK may have limited direct motivational force unless activated through social-normative and feasibility-based mechanisms. This dual-process pattern implies that cognitive information (EK) must first be channeled through social reference groups (SN) and perceived capability (PBC) before it can influence behavioral intentions—a finding consistent with dual-process theories of persuasion (Petty & Cacioppo, 1986). Additionally, the finding that EC retains a weak direct effect while EK does not may indicate that affective-moral engagement (EC) maintains a residual direct motivational link to intention that bypasses the modeled mediators. Furthermore, since knowledge is seen as a prerequisite to social sensitivity, the present research adds to the use of the TPB by demonstrating the ability of cognitive factors (EK) to trigger the influence of norms (SN). This adds to a better synthesis of how both internal and social psychological processes combine to influence green intentions.

More broadly, it is important to acknowledge that our model does not capture the full range of theoretically relevant mechanisms that may carry EC-to-GPI effects. The Norm Activation Model (Schwartz, 1977) identifies personal moral norms as a distinct pathway from EC to pro-environmental behavior—one that is conceptually separate from both SN (externally referenced social pressure) and PBC (perceived feasibility). The absence of personal norms as a mediator represents a theoretical gap in the present model, and the residual direct effect of EC on GPI may partly operate through this unmeasured moral obligation pathway. Additionally, other TPB-adjacent constructs—including affect, self-identity, and habit—that have demonstrated explanatory power in green consumption research are not captured in the present framework. This represents a meaningful theoretical limitation beyond the attitude omission discussed earlier, and future research should consider multi-mediator models that include these mechanisms alongside the present ones to provide a more comprehensive account of how environmental orientations are translated into GPI.

This interpretation aligns with dual-process frameworks in social psychology, which posit that cognitive inputs (such as factual knowledge) are motivationally inert until they are channeled through affective, normative, or identity-based pathways that are more proximally connected to behavioral decisions (Chaiken & Trope, 1999; Petty & Cacioppo, 1986). In the present model, EK requires activation through SN (social reference) and PBC (perceived capability) before it can generate GPI, suggesting that environmental education programs may be more effective when they simultaneously strengthen social norms around green consumption and enhance consumers' PBC, rather than relying on knowledge dissemination alone.

From a managerial perspective, the findings suggest that increasing GPI in China requires more than simply raising awareness of

environmental issues. Because PBC is the strongest mediating pathway, policymakers and marketers should reduce practical barriers to green purchasing by improving product availability, offering credible eco-labels, reducing information asymmetry, and supporting affordability through incentives or price-related interventions. The significance of SN further indicates that social influence strategies are likely to be effective. Campaigns that use peer endorsement, community leaders, influencers, and socially visible green norms may strengthen consumers' intentions by framing green purchasing as a socially supported behavior. Finally, the non-significant GPV mediation result suggests that value appeals alone may not be sufficient to translate awareness into intention. Marketers should therefore combine value-based messaging with concrete enablers such as accessibility, trust-building, and reductions in perceived risk. Taken together, these findings imply that successful green consumption strategies should address not only environmental awareness, but also consumers' social environment and perceived behavioral capability.

6. Conclusion

This study examined how EK and EC shape GPI among Chinese consumers through three proposed mediating mechanisms: SN, PBC, and GPV. The findings show that both EK and EC are important antecedents of GPI, but they operate through different pathways. EK influences GPI indirectly, whereas EC exerts both a weak direct effect and significant indirect effects. In both cases, SN and especially PBC emerge as the principal mediating mechanisms, while GPV does not function as a significant mediator in this model.

These findings suggest that GPI in China is shaped less by value appraisal alone and more by whether consumers perceive green purchasing as socially supported and behaviorally feasible. The study therefore contributes to the literature by comparatively examining multiple mediating pathways within a TPB-informed integrated model rather than treating environmental antecedents as only direct predictors of intention. The study also has limitations. The use of cross-sectional data prevents strong causal inference, and the sample is heavily concentrated among young and highly educated respondents (predominantly aged 19–23), which limits generalizability. This demographic composition is theoretically consequential: younger, highly educated consumers are particularly susceptible to peer influence, digitally mediated normative messaging, and sustainability discourse propagated through social media. Consequently, the strong effects of SN and PBC in this study may reflect, at least in part, the elevated normative sensitivity of this subpopulation rather than universal consumer behavior patterns. Future research should replicate the model across broader age cohorts, income levels, and geographic regions to determine whether the observed mediating pathway pattern generalizes beyond this educational and demographic context. In addition, attitude, a canonical TPB construct, was not modeled separately from GPV. Future research should therefore examine attitude and GPV side by side and test the model across broader demographic groups and additional consumption contexts.

More specifically, the overwhelming concentration of respondents in the 19–23 age cohort (88.8%) represents not merely a generalizability constraint but a theoretically consequential scope limitation. This demographic is specifically characterized by: (1) heightened susceptibility to peer influence and social conformity pressures, consistent with established findings in social influence research (Cialdini & Goldstein, 2004); (2) intensive engagement with social media platforms through which sustainability norms are amplified, making digitally mediated normative influence a particularly strong predictor for this cohort; and (3) limited financial autonomy as university students, which may suppress value-for-money assessments and reduce the explanatory power of GPV. We therefore explicitly acknowledge that the stronger mediating roles of SN and PBC in our findings may partially reflect the motivational profile of young, socially networked, digitally embedded consumers, rather than Chinese green consumers in general. Future research should replicate the model across broader age cohorts, lower-educated groups, income segments, and consumers from less digitally connected regions of China, preferably using quota sampling or stratified probability sampling to assess whether the observed mediating pathway pattern generalizes beyond young, highly educated respondents.

CRediT authorship contribution statement

Zhang Yiping: Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ratneswary Rasiah:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization. **Song Lin:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation. **Shahryar Sorooshian:** Writing – review & editing, Writing – original draft, Conceptualization.

Ethical approval

While formal institutional ethical approval was waived due to the minimal risk nature of the study, the research was conducted under the direct supervision of SEGi University, as a PhD thesis.

Consent for publication

All authors consent to the publication of this article and affirm that the manuscript represents original work not under consideration elsewhere.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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